Optimization Methods with External Mesh (Code_v2)

Fachseminar Report

January 8, 2026

Overview

This report summarizes the Ex3 reduced optimal control problem using an external mesh with a circular control region.

1 Problem and Discretization

We consider $\Omega = (0,1)^2$, y solving $-\Delta y = \chi_\omega u$ with $y = 0$ on $\partial\Omega$, and objective $F(u) = \frac{1}{2}\|y - y_d\|^2 + \frac{\beta}{2}\|u\|^2$. Finite elements: CG1 for state/control; operators assembled from XDMF subdomain tags.

2 Optimization Methods

Barzilai–Borwein, fixed-step GD with $\alpha = 1/L$, and Nesterov acceleration are applied in the control space with inner product induced by M_U.

3 Figures

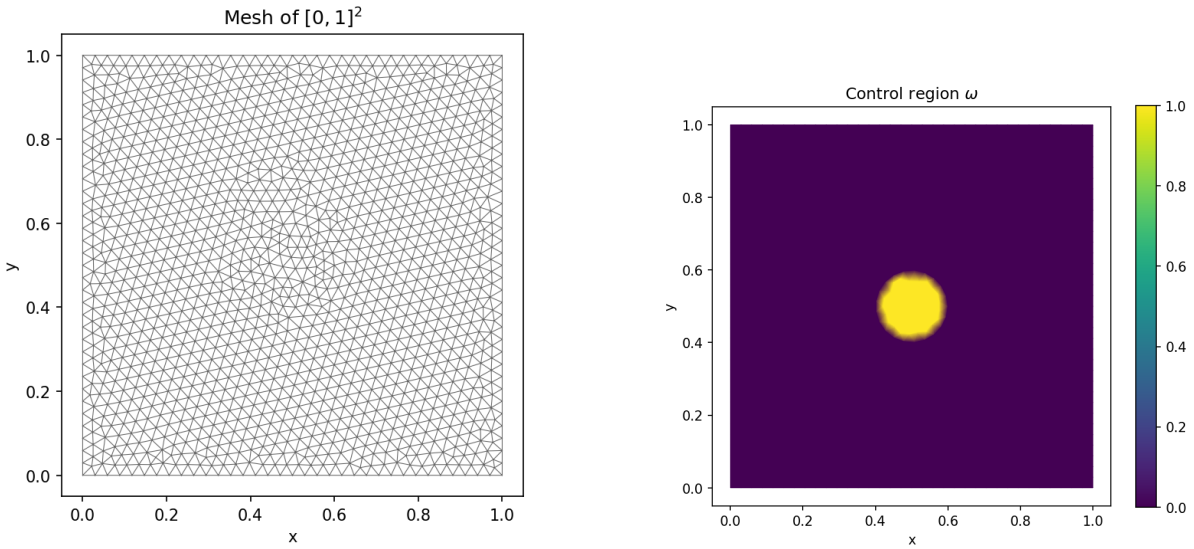

Figure 1: Mesh and circular ω.

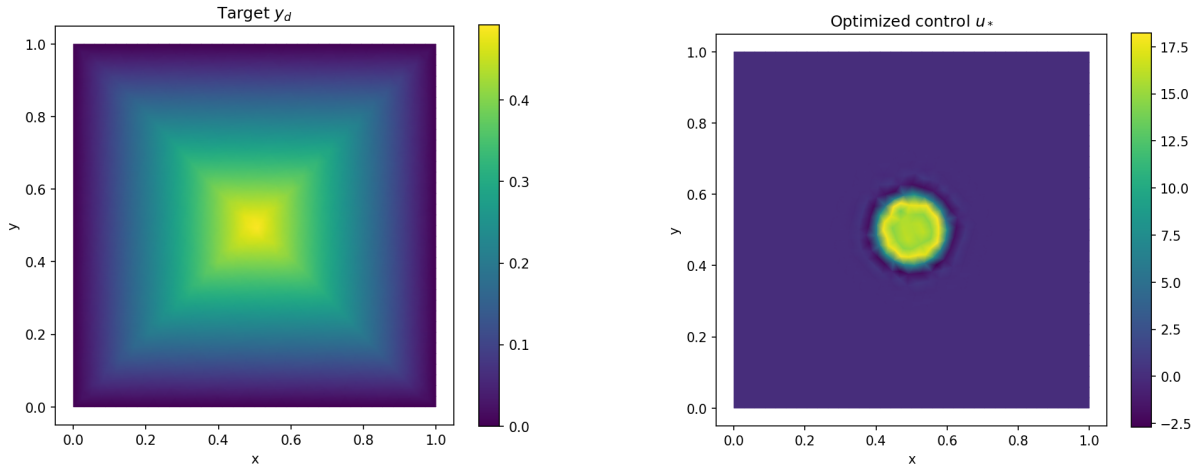

Figure 2: Target y_d and optimized control u_*.

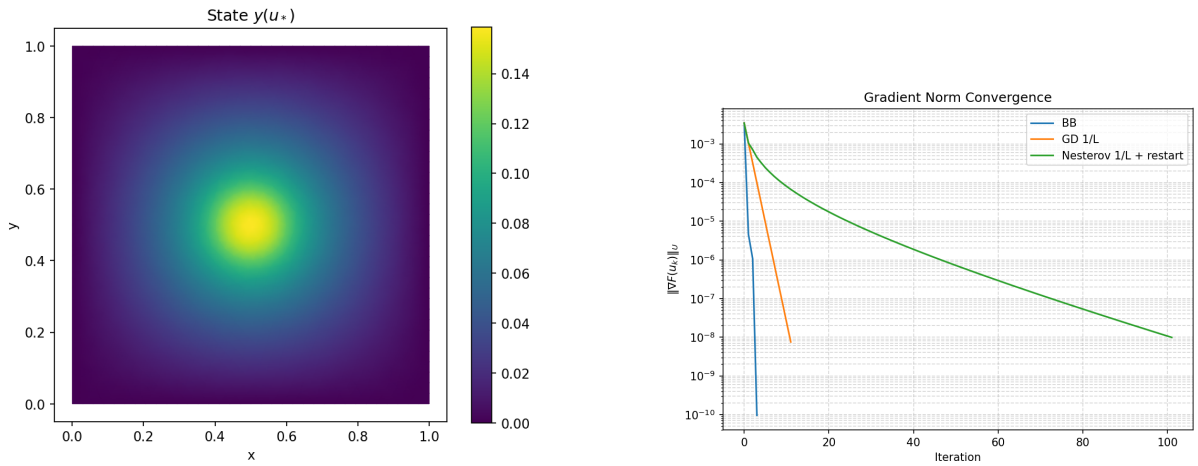

Figure 3: State $y(u_*)$ and gradient norm convergence.

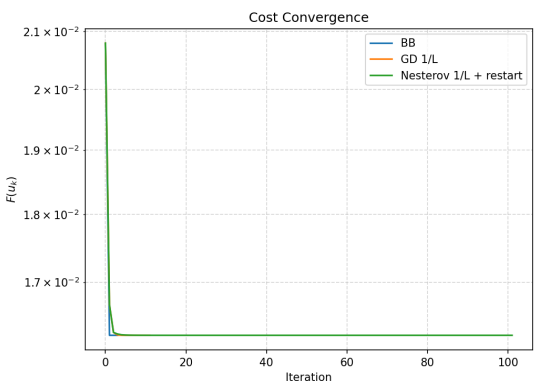

Figure 4: Objective convergence.